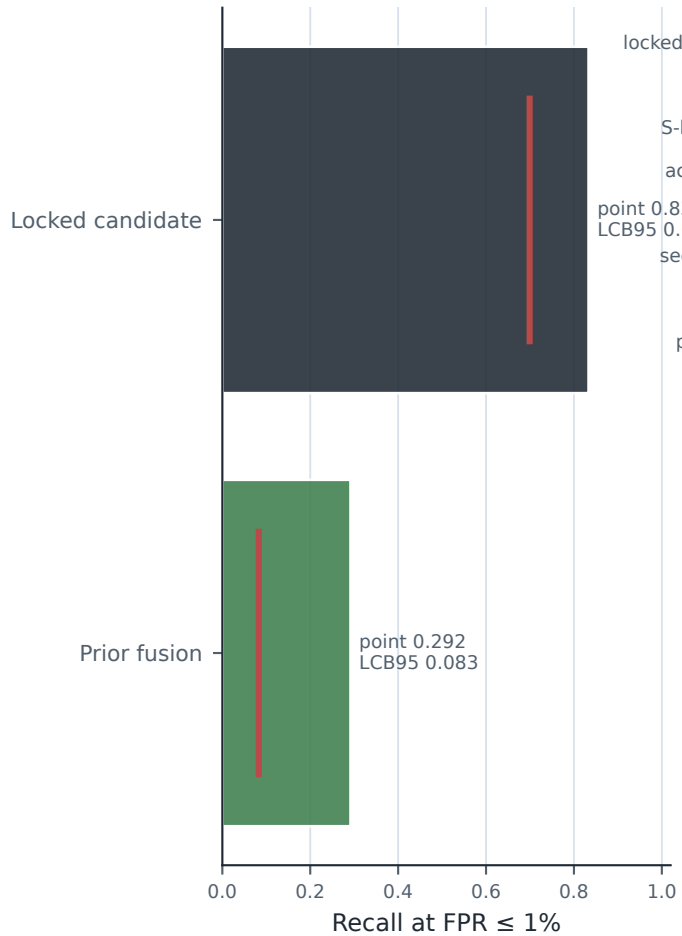
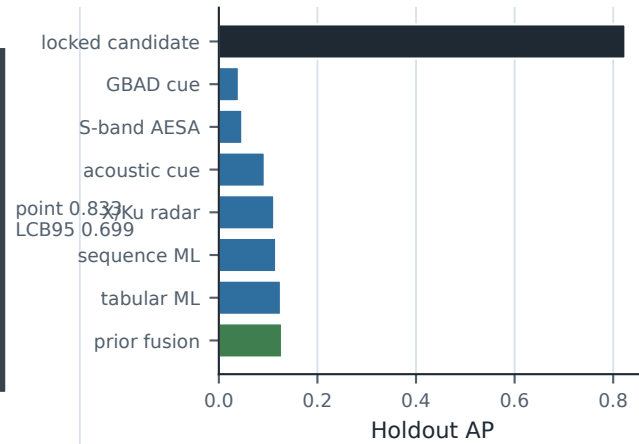


Holdout primary KPI, ranking diagnostics, low-FPR curves, and calibration

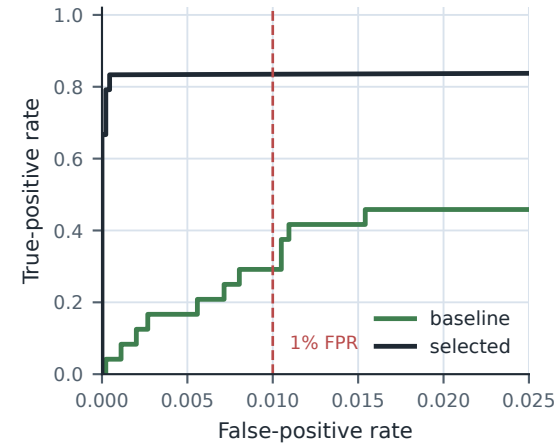
Primary KPI: group-block LCB95



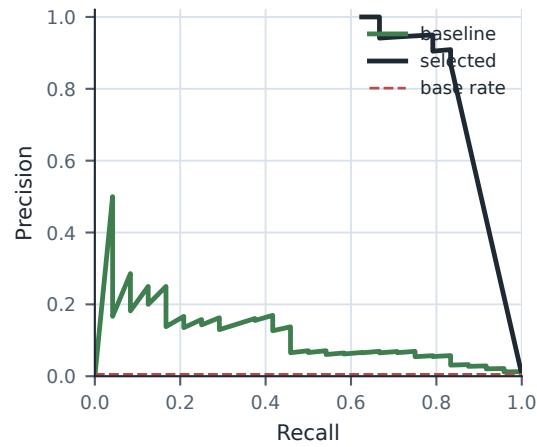
AP ranking diagnostic



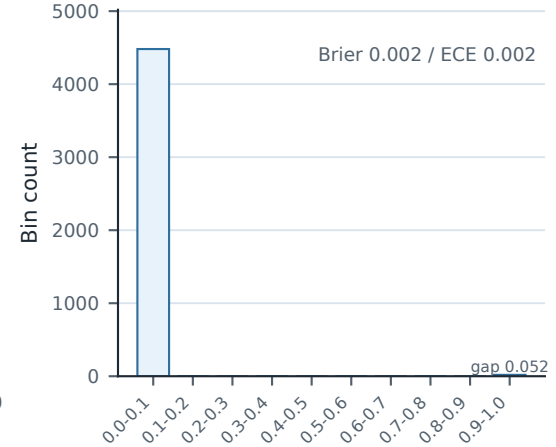
ROC low-FPR inset



PR curve



Calibration bin counts



LCB95 Recall@≤1%FPR: 0.699; holdout has 8 positive groups / 24 positive records.

AP and ROC AUC are supporting rank diagnostics; selected ROC AUC is lower than prior fusion.